

DECODELABS TASK 01 - PASSWORD CHECKER USING PYTHON

CODE:

```
# Password Strength Checker
# DecodeLabs Cyber Security Project 1

password = input("Enter your password: ")

# Character checks
has_upper = False
has_lower = False
has_digit = False
has_symbol = False

symbols = "!@#$%^&*()-_+=<>?/"

for char in password:
    if char.isupper():
        has_upper = True
    elif char.islower():
        has_lower = True
    elif char.isdigit():
        has_digit = True
    elif char in symbols:
        has_symbol = True

# Strength calculation
score = 0

if len(password) >= 8:
    score += 1

if has_upper:
    score += 1
```

```
if has_lower:
    score += 1
if has_digit:
    score += 1
if has_symbol:
    score += 1

# Result
if score <= 2:
    strength = "Weak"
elif score <= 4:
    strength = "Medium"
else:
    strength = "Strong"
print("\nPassword Strength:", strength)

# Extra security suggestions
if not has_upper:
    print("Add uppercase letters")
if not has_digit:
    print("Add numbers")
if not has_symbol:
    print("Add symbols")
if len(password) < 8:
    print("Use minimum 8 characters")
```

OUTPUT:

- PS D:\DecodeLabs> python -u "d:\DecodeLabs\Task1_Password_Checker.py"
Enter your password: Hello123

Password Strength: Medium
Add symbols

- PS D:\DecodeLabs> python -u "d:\DecodeLabs\Task1_Password_Checker.py"
Enter your password: Hello123_21

Password Strength: Strong

- PS D:\DecodeLabs> python -u "d:\DecodeLabs\Task1_Password_Checker.py"
Enter your password: Hello

Password Strength: Weak
Add numbers
Add symbols
Use minimum 8 characters

PS D:\DecodeLabs> █